# AIRLINE PASSENGER SATISFACTION ANALYSIS

AQUILA, KRISHNA

# INTRODUCTION

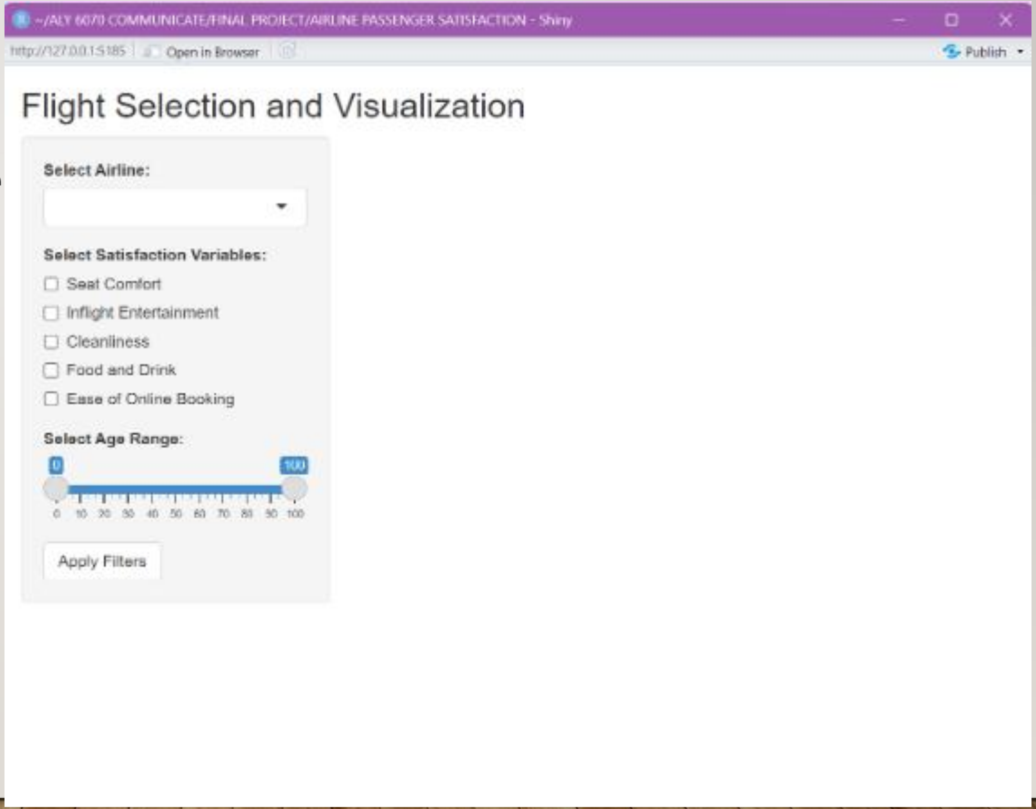
In today's fast-paced aviation landscape, the pursuit of passenger satisfaction stands as a cornerstone for airlines aiming to excel in service delivery and maintain a competitive edge.

In this analysis, we leveraged various platforms such as Shiny R, Tableau, and Power BI.

The Airline Passenger Satisfaction dataset was sourced from Kaggle.

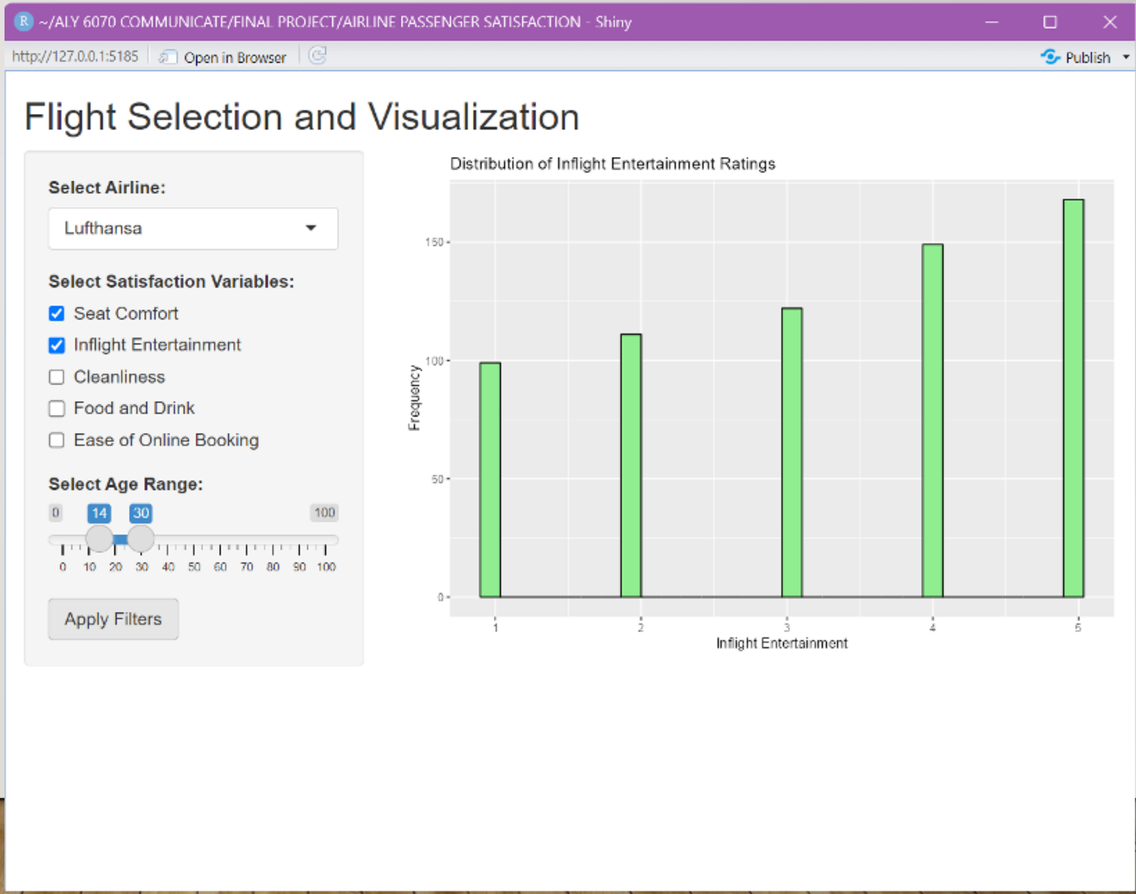
# SHINY R

Tab showing comparison by certain satisfaction variables like age, airline, and other factors.



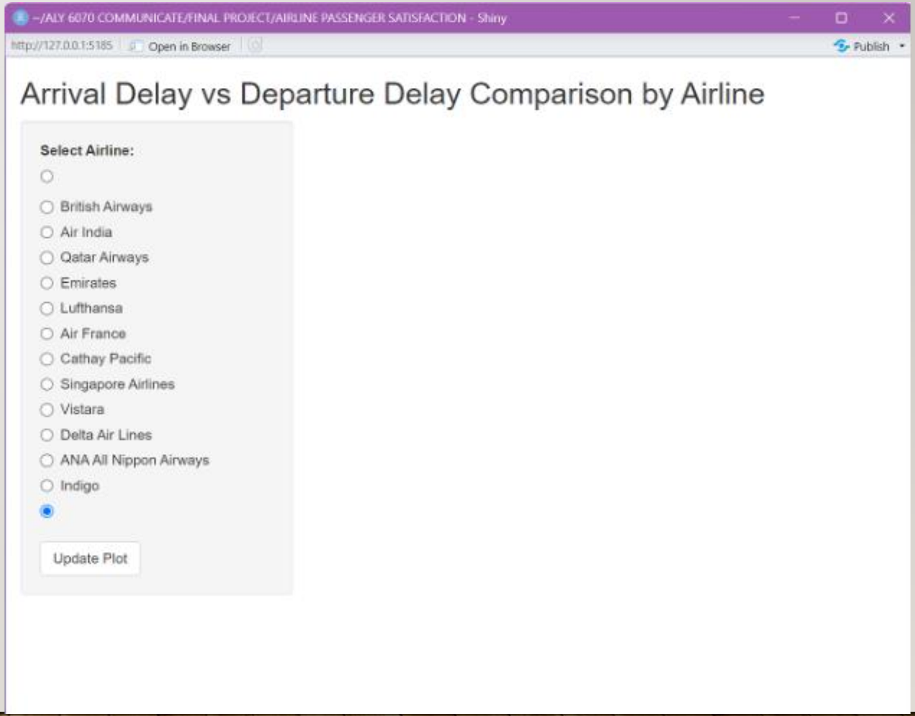
# SHINY R

After selecting certain factors



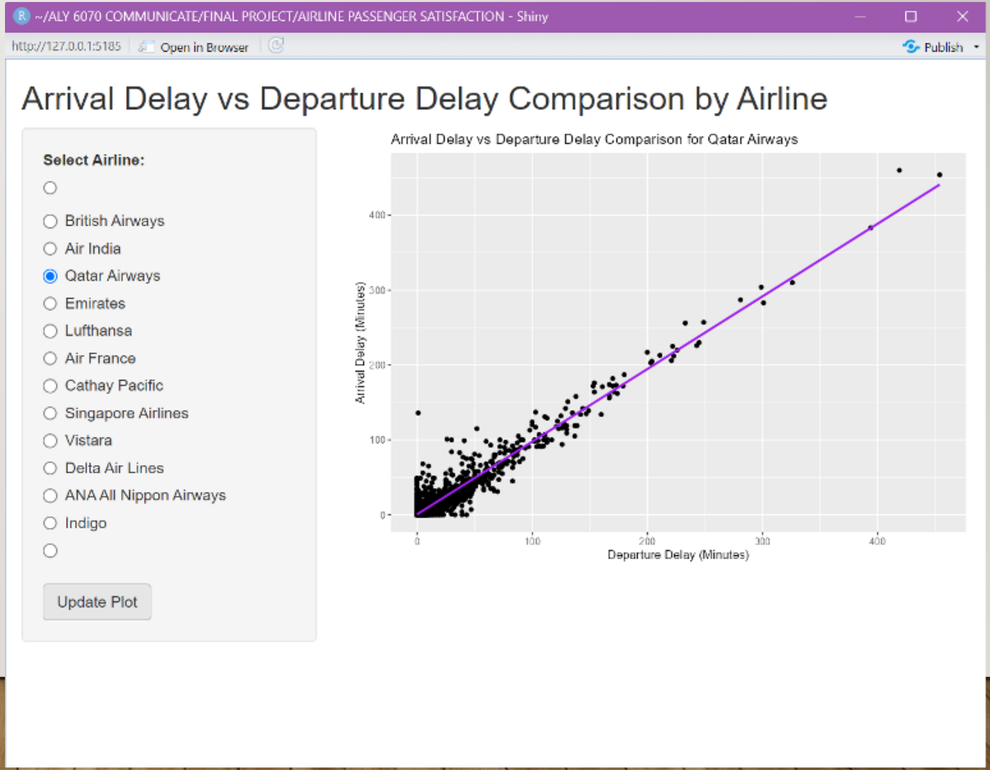
# SHINY R

Tab showing comparison of arrival and departure delay by airline name.



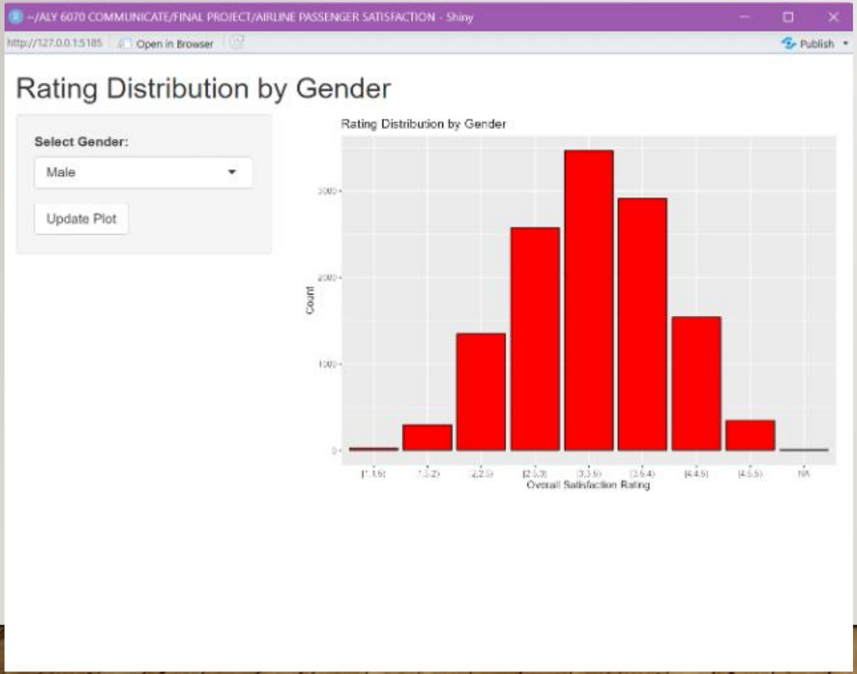
# SHINY R

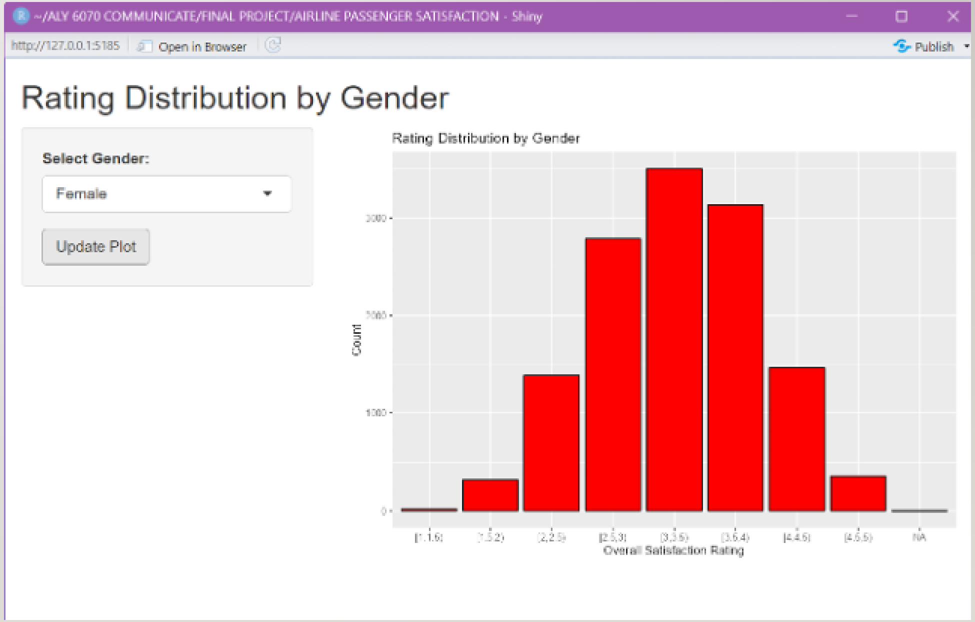
After selecting a certain Airlines



# SHINY R

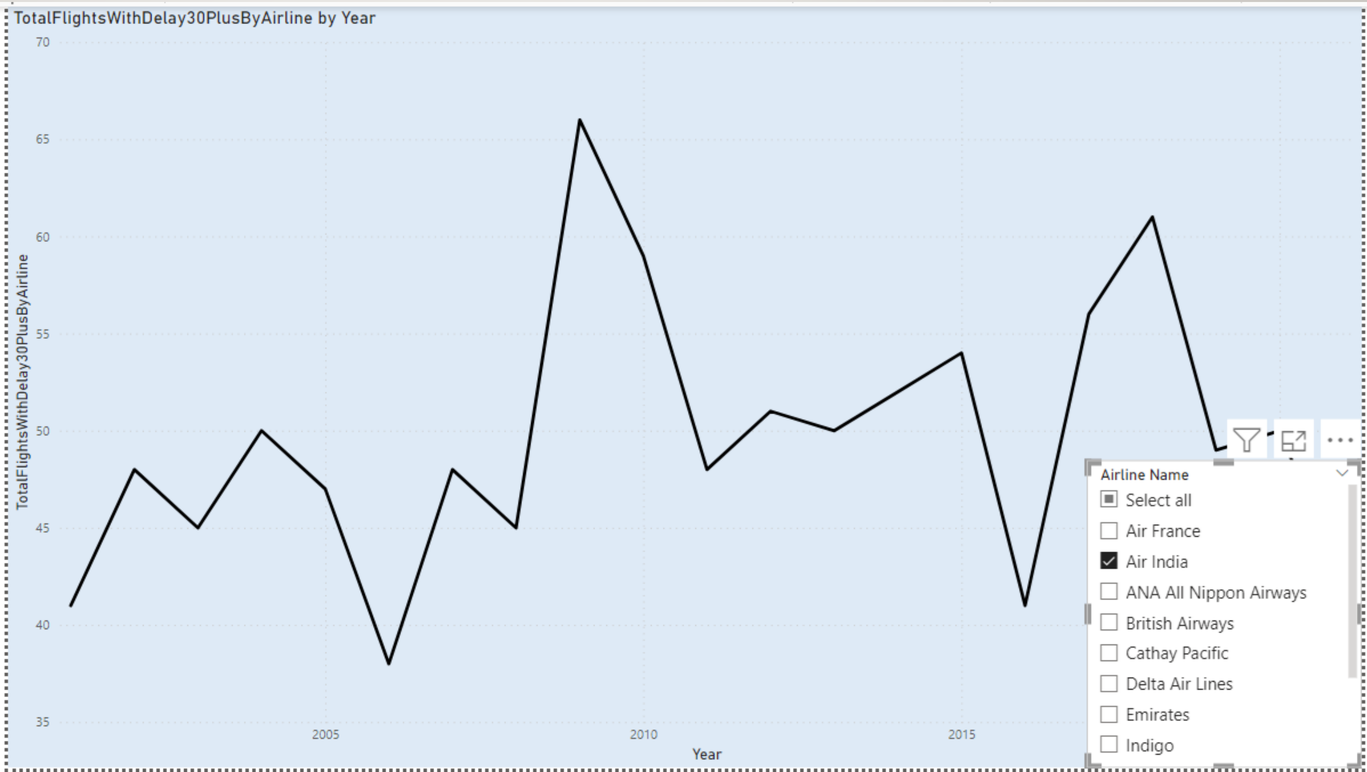
Rating Distribution by gender



# POWERBI

Line chart showing total flights with delays greater than 30 minutes over the years.



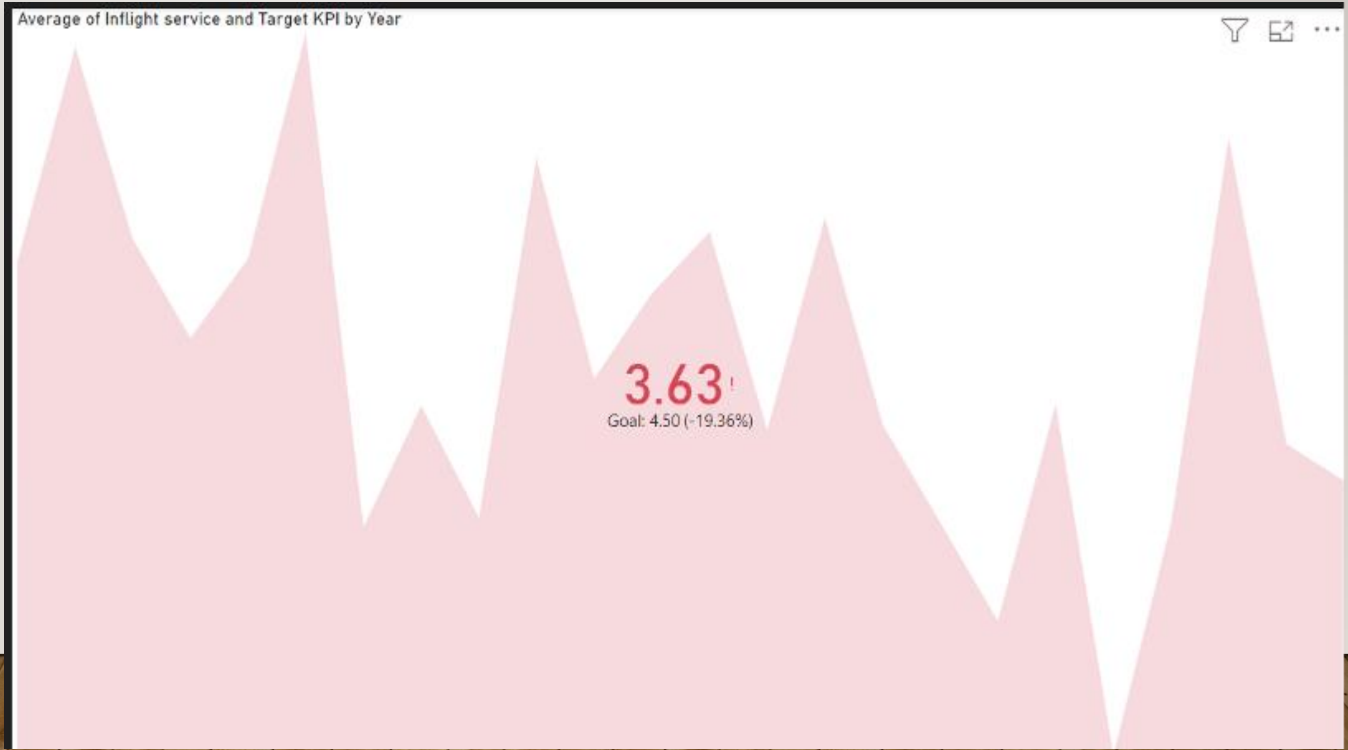
# POWERBI

DAX formula which compares the total flights with delays of more than 30 minutes, filtered by airline name.

```
TotalFlightsWithDelay30PlusByAirline =
SUMX(
    VALUES('Airline Passenger Satisfaction'[Airline Name]),
    CALCULATE(
        COUNTROWS('Airline Passenger Satisfaction'),
        'Airline Passenger Satisfaction'[Departure Delay in Minutes] > 30
    )
)
```

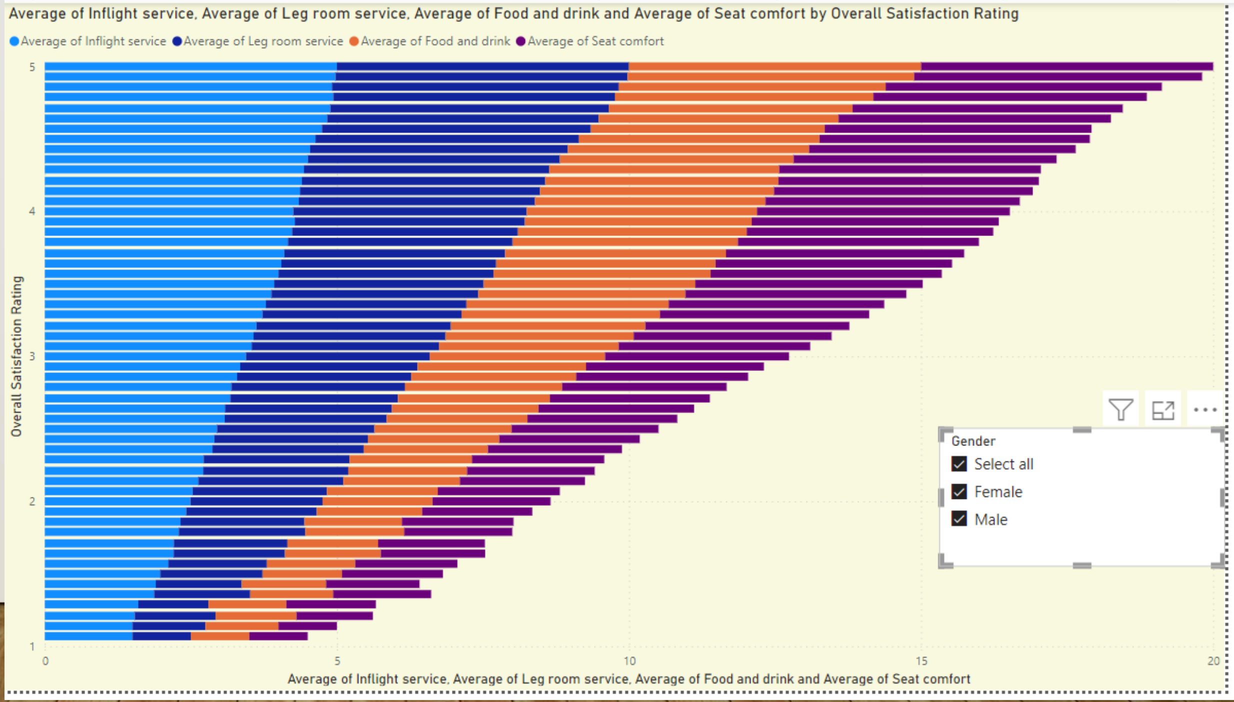
# POWERBI

KPI of average of in-flight services by year



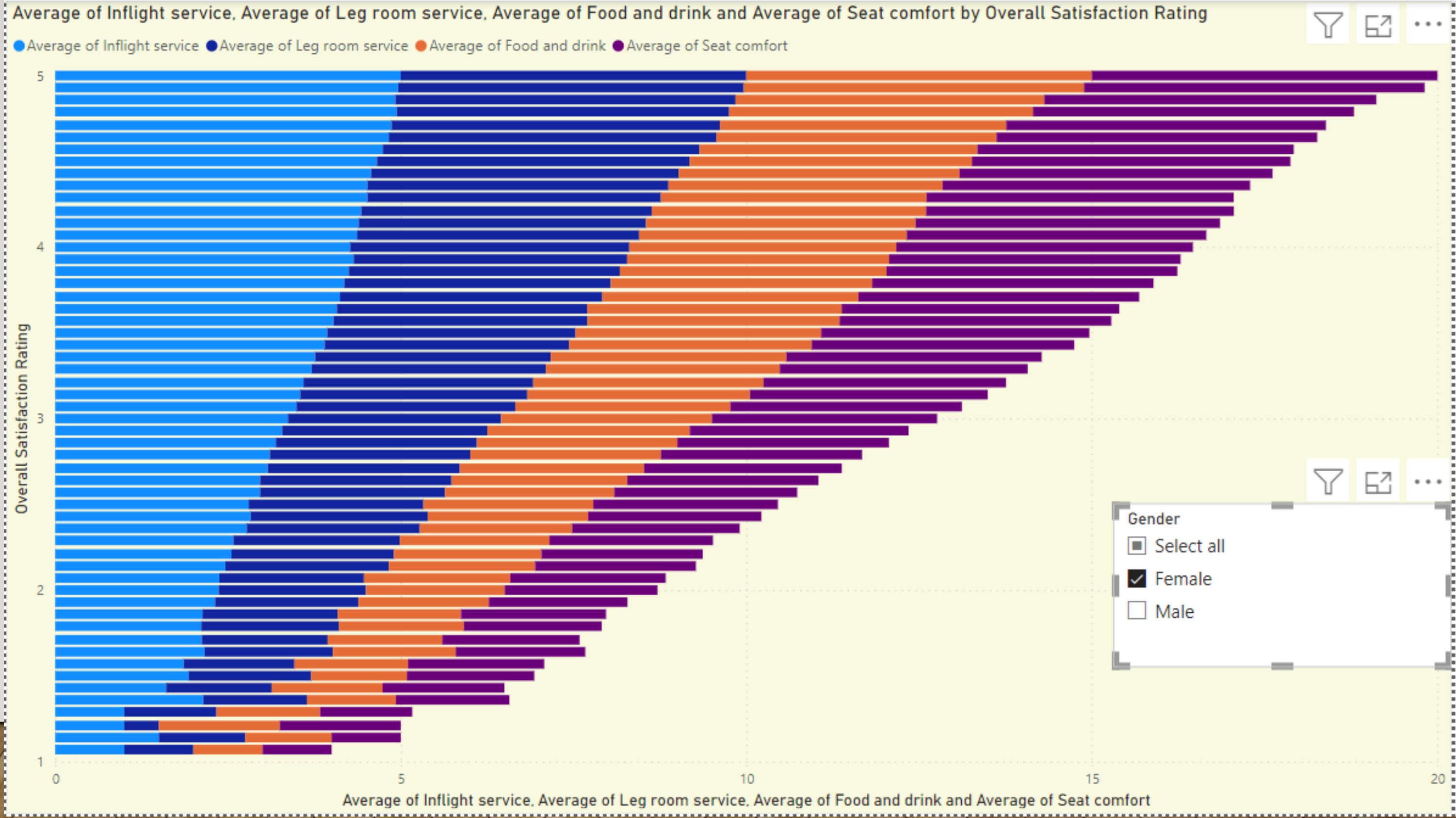
# POWERBI

Bar Chart of Average of Certain Factors That Influence Passenger Satisfaction by Gender



# POWERBI

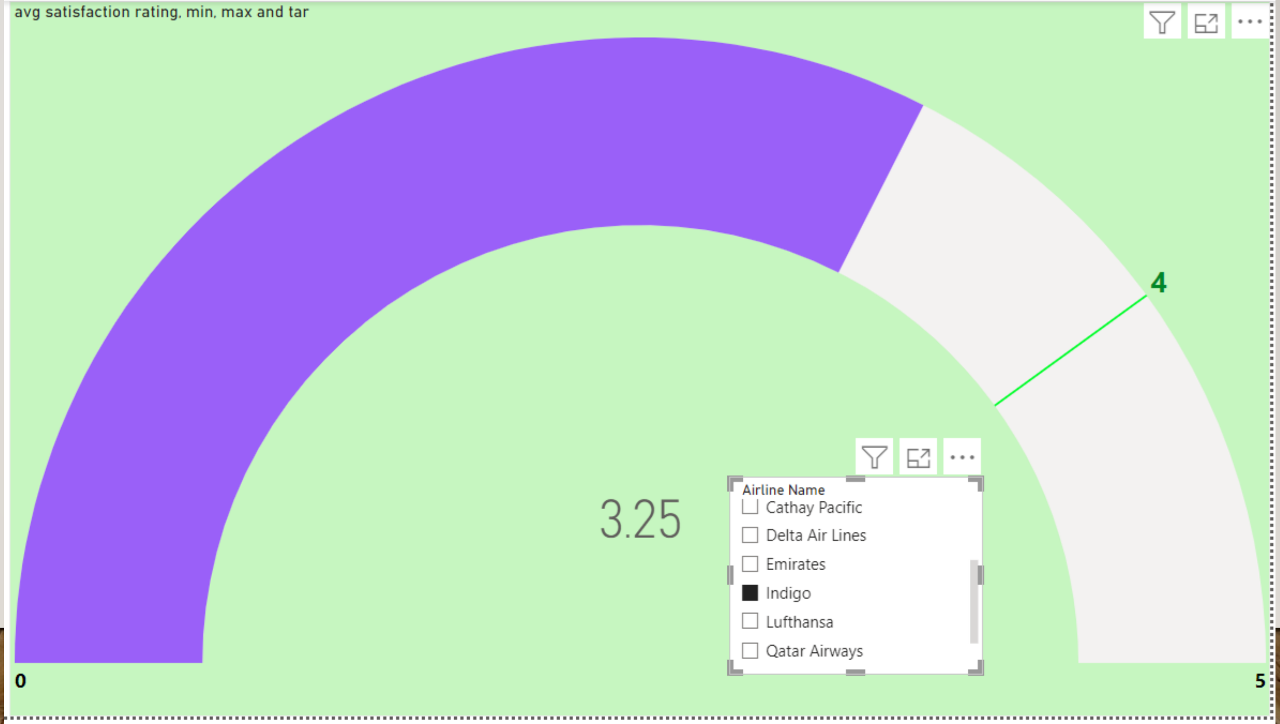
Filter Applied on Female selection.



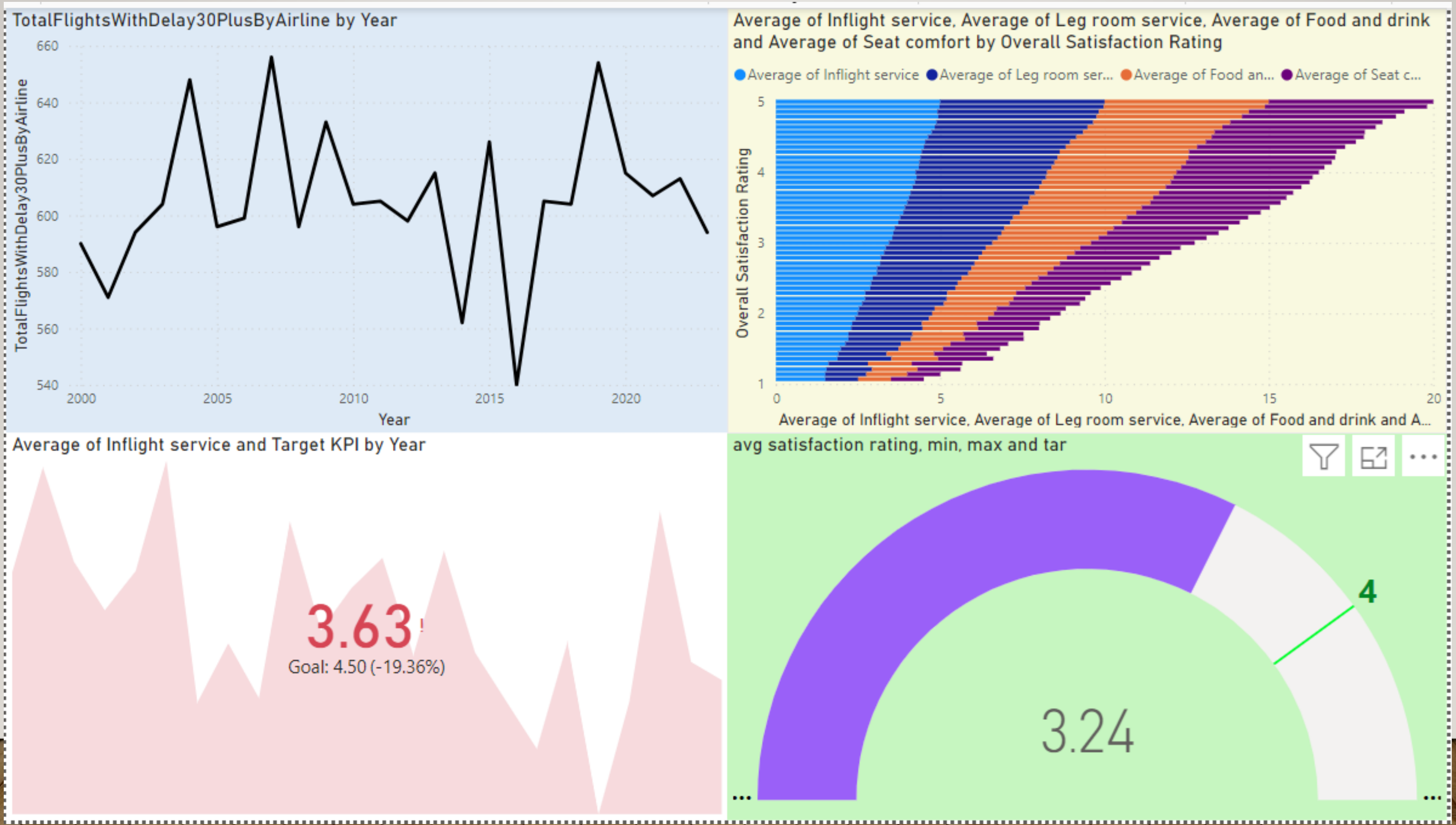
# POWERBI

Gauge chart for overall satisfaction rating with airline filter.

The target value is 4

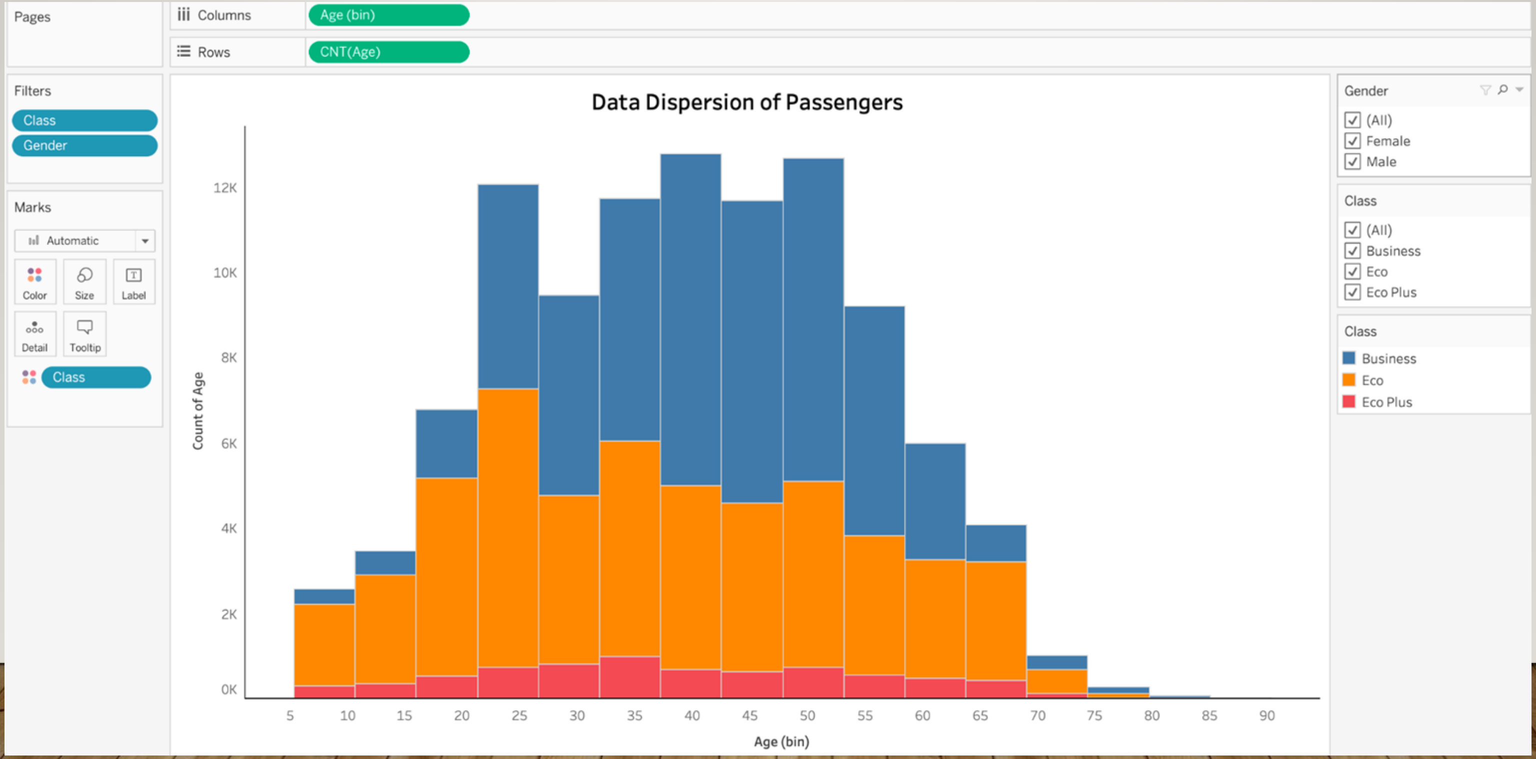


# POWERBI DASHBOARD

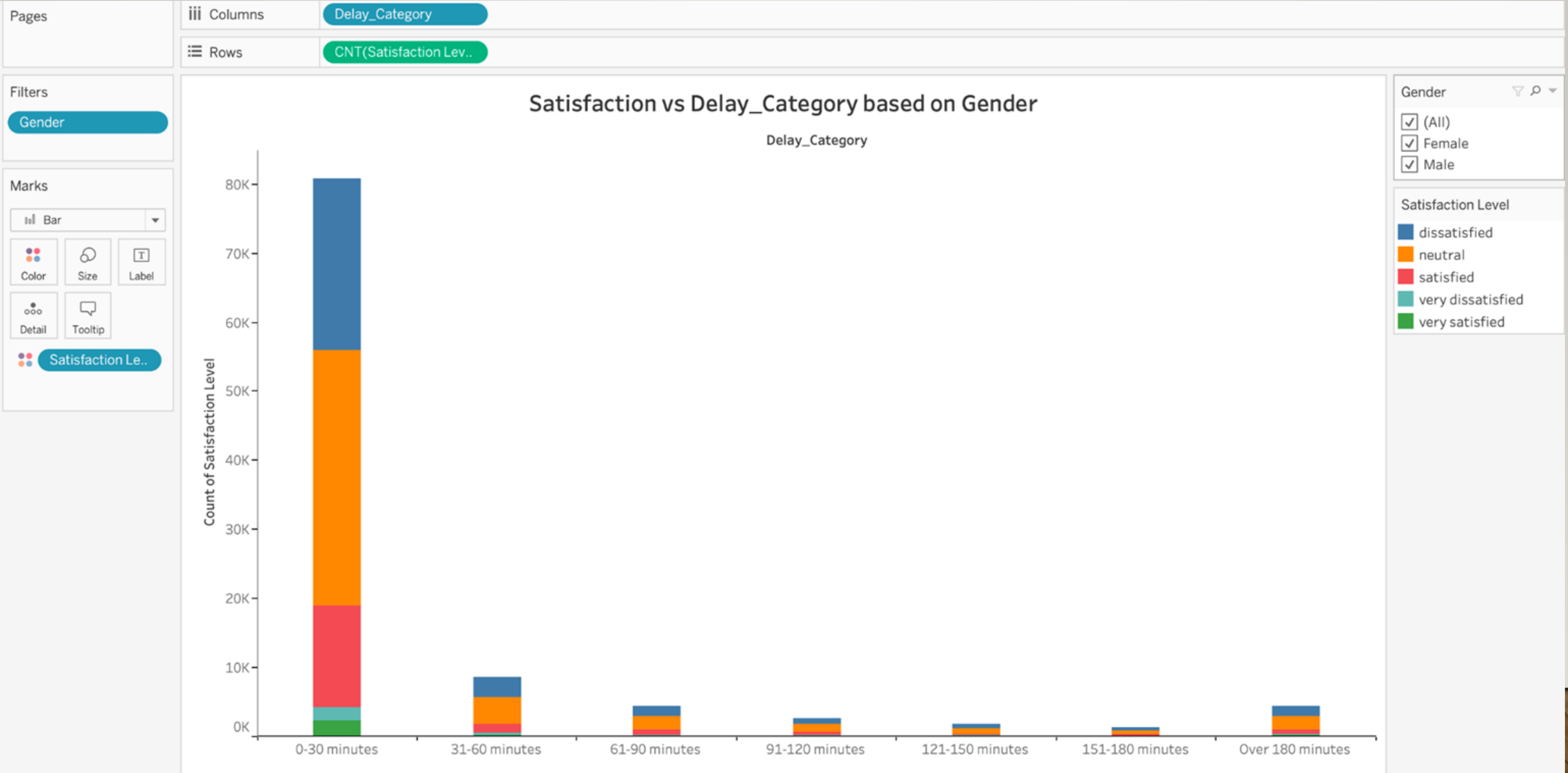


# TABLEAU

**Data Dispersion of Passengers**

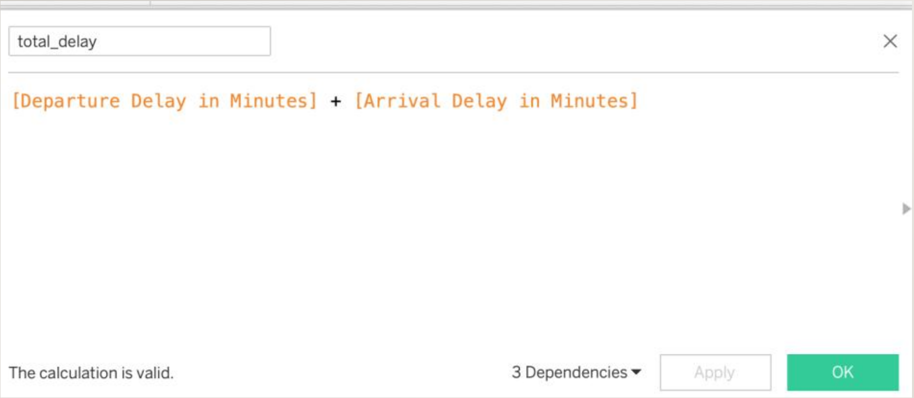


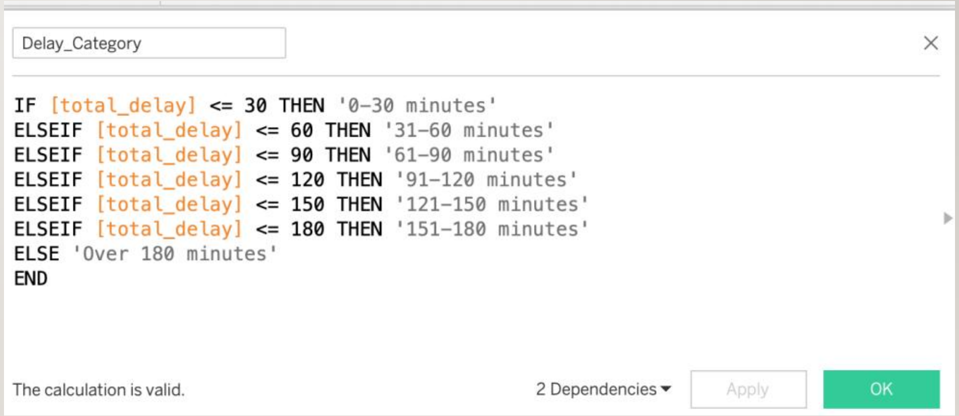
## Satisfaction vs Delay graph based on gender



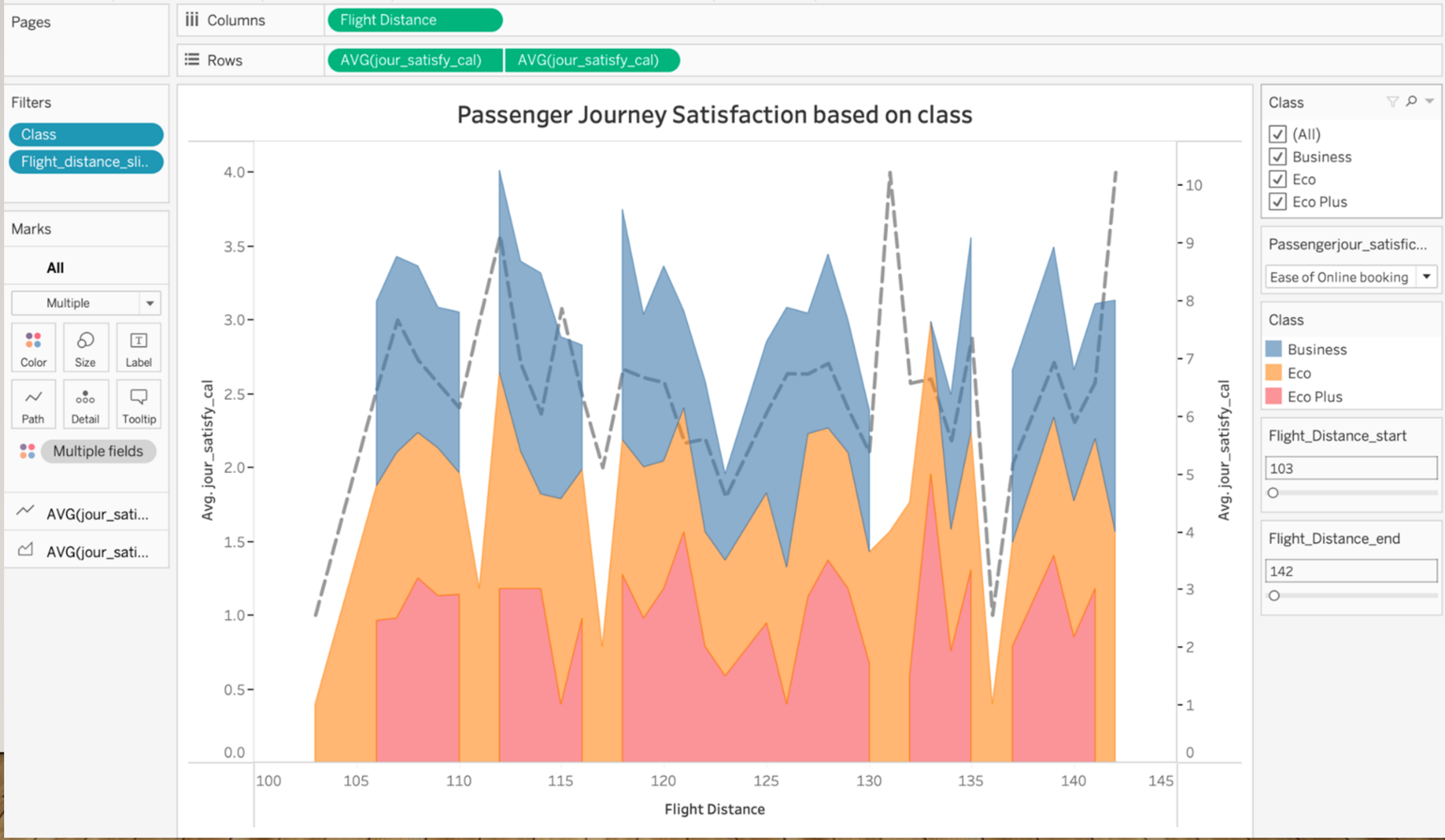
# TABLEAU

Created two calculated fields called Total_delay and Delay_category.



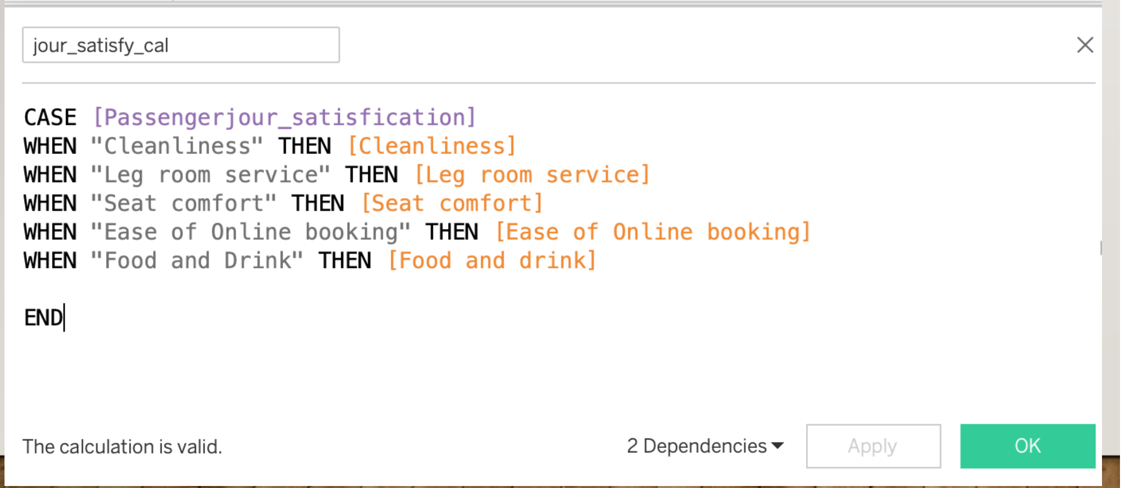



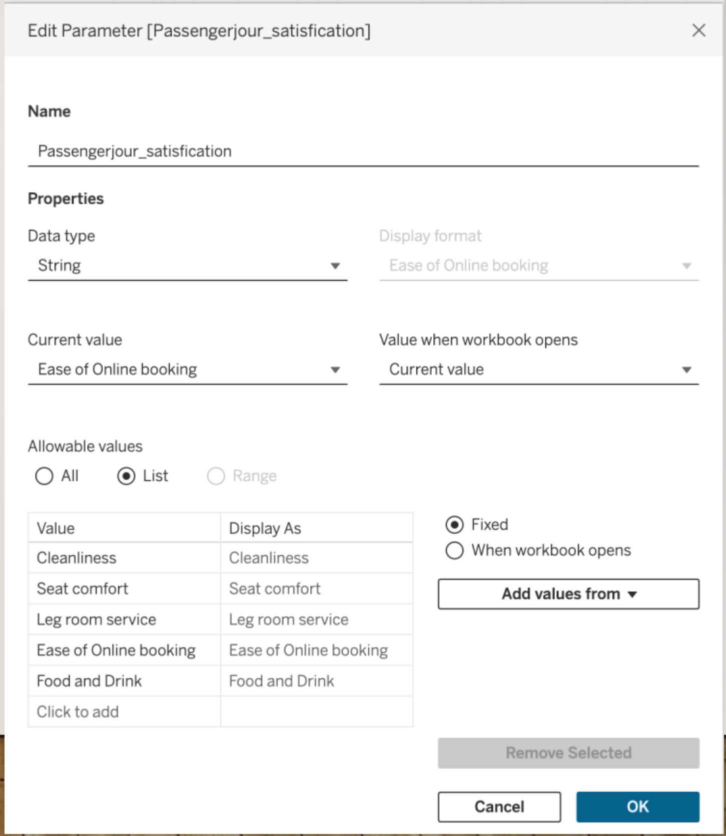
## Passenger Journey Satisfaction graph



# TABLEAU

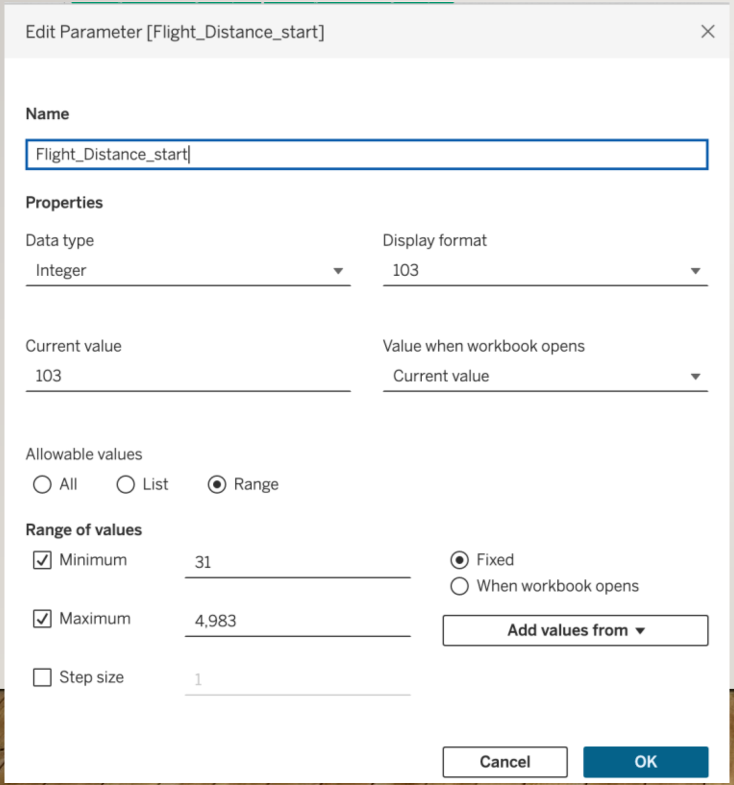
For this graph, created a jour_satisfy_cal (parameter), Passengerjour_satifaction (calculated field) and flight_distance slider.

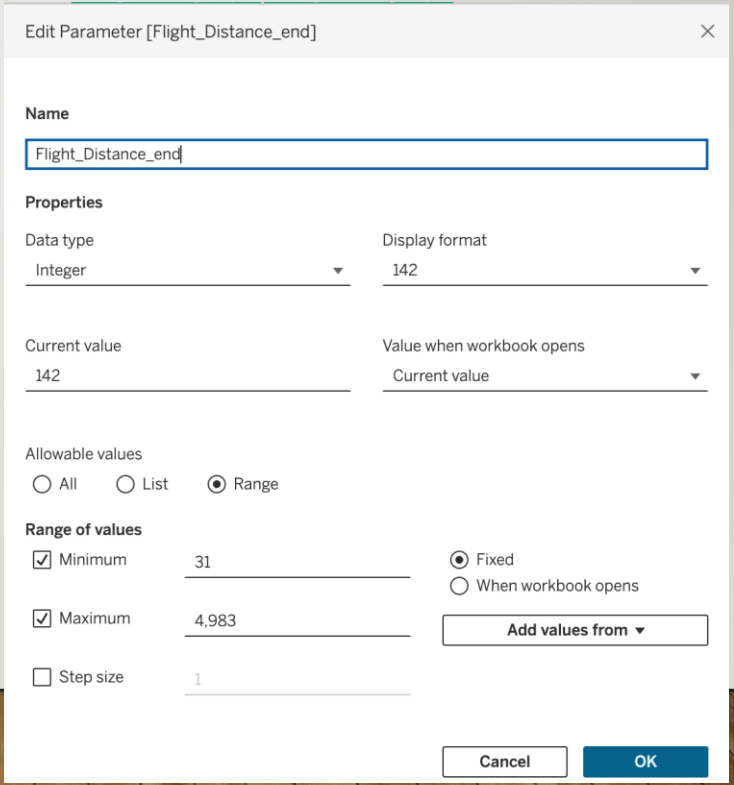


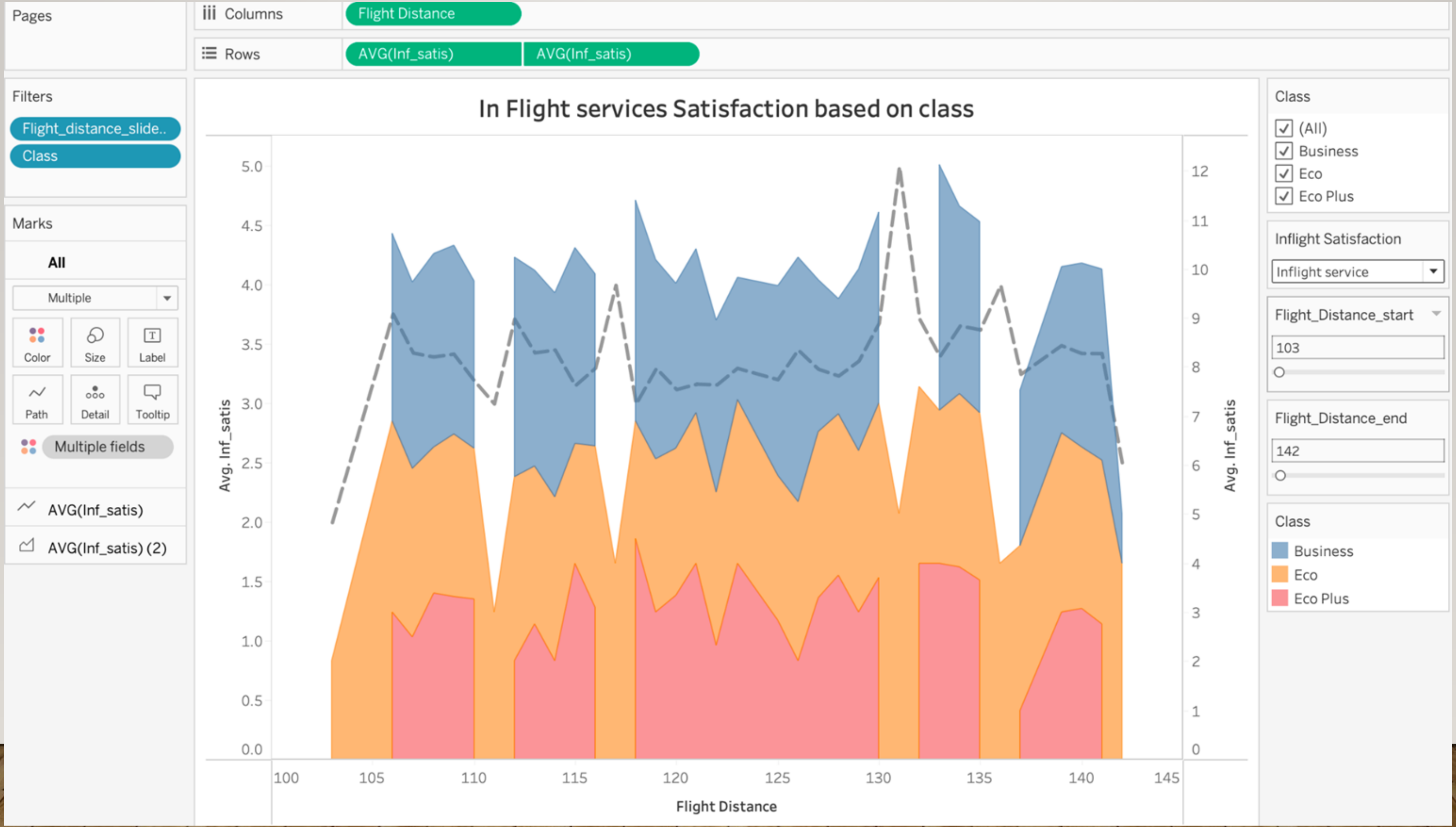
# TABLEAU

Flight_distance Sliders



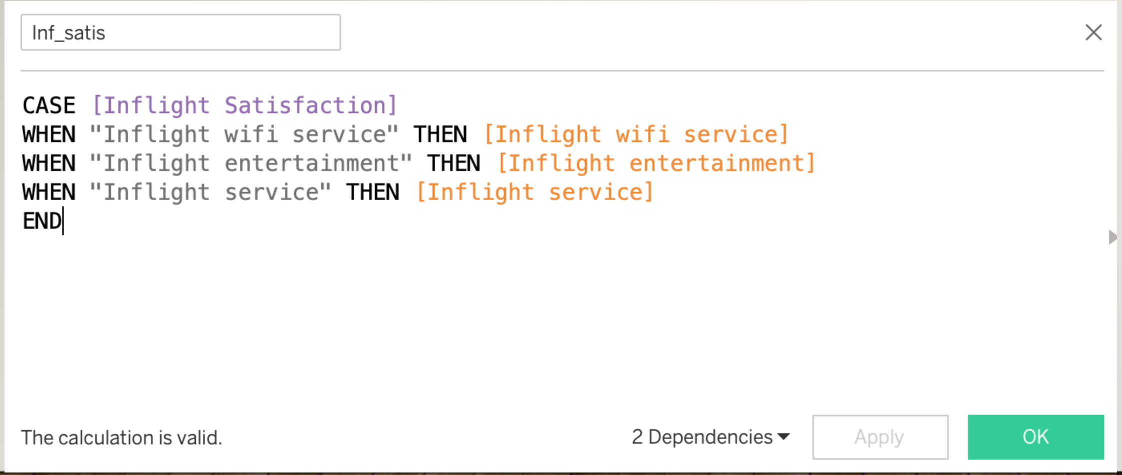



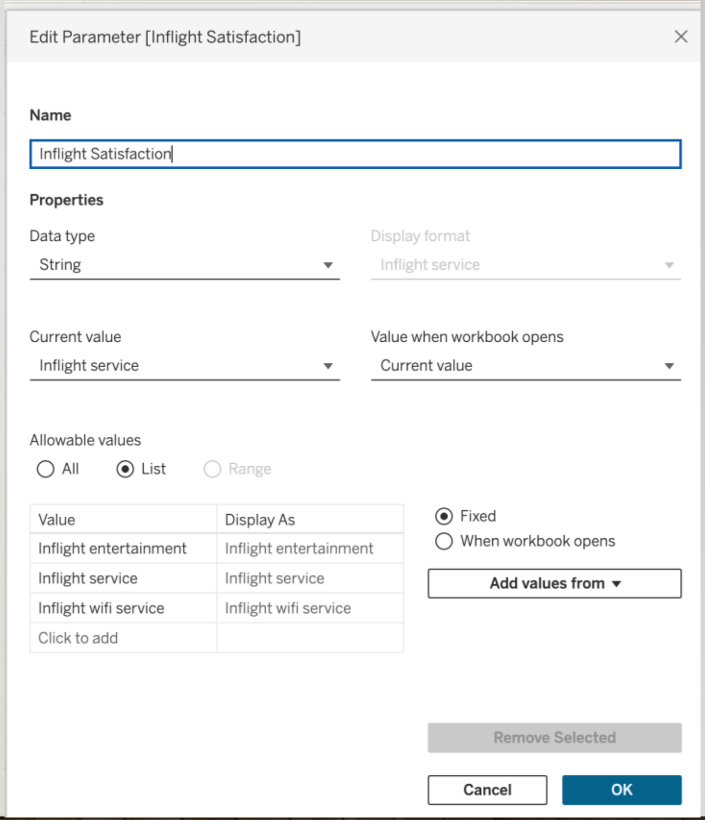
## In Flight Service Satisfaction graph



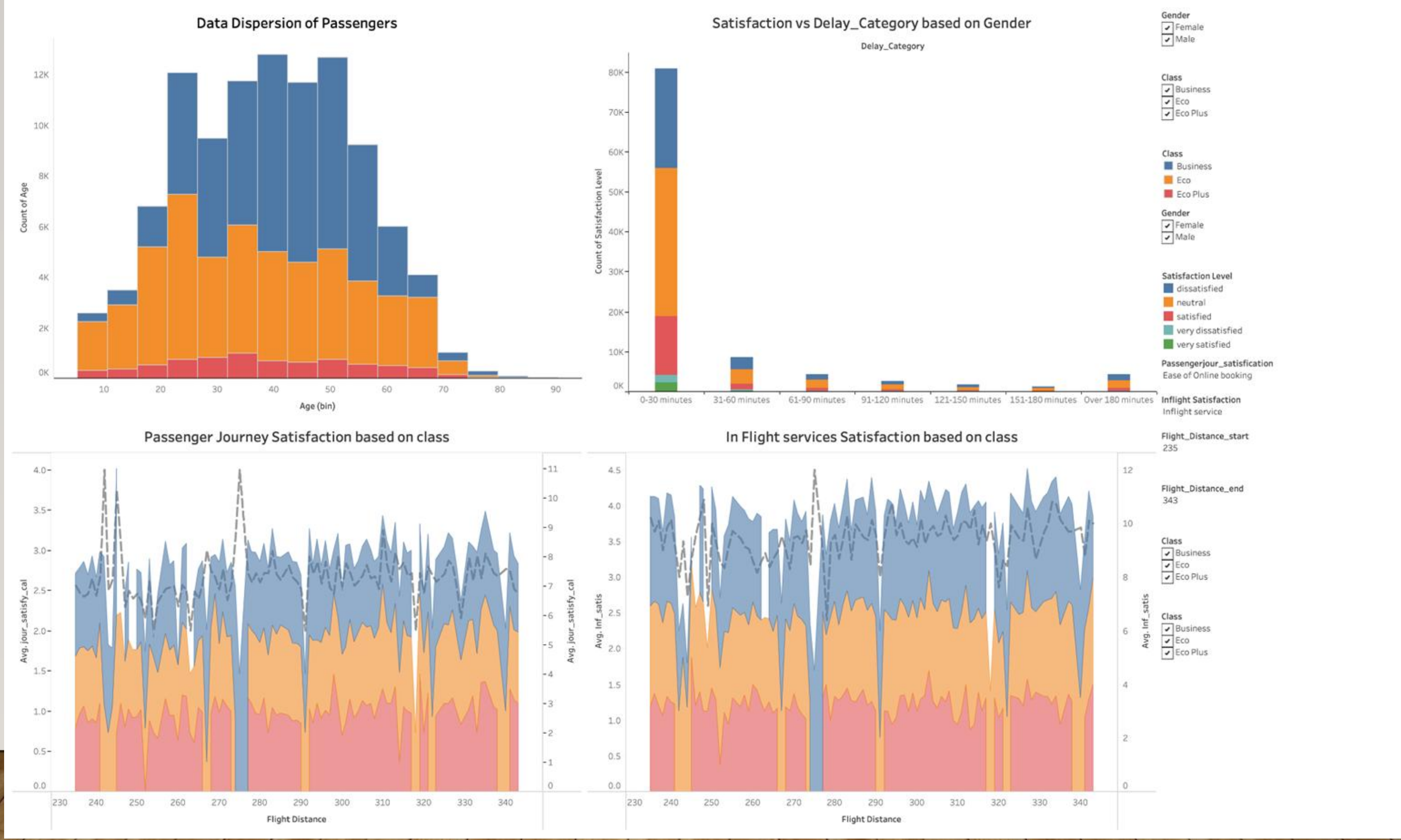
# TABLEAU

For this graph, created a Inf_satifs(parameter), and Inflight Satisfaction (calculated field).



# TABLEAU DASHBOARD



# CONCLUSION

The study conducted a thorough analysis of airline passenger satisfaction, employing tools such as Shiny R, Tableau, and Power BI for a deep dive into the data.

Through interactive visualizations, the study was able to uncover critical trends and correlations that affect passenger experiences in the aviation industry

# REFERENCES

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